MIL-L-43582B
19 June 1981
SUPERSEDING
MIL-L-43582A
5 April 1978

MILITARY SPECIFICATION

LINER, SHIRT, CHEMICAL PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a cotton sateen shirt liner.

1.2 Classification. The liner shall be of one type in the following sizes (see 6.2).

Schedule of sizes

X-small
Small
Medium
Large
X-large

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

FEDERAL

V-B-871	-	Button, Sewing Hole, and Button Staple (Plastic)
V-T-276	-	Thread, Cotton
V-T-280	-	Thread, Gimp, Cotton, Buttonhole
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-10296	-	Cloth, Sateen, Cotton

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in accordance with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the specification may appear in the sample, in which case this specification shall govern.

3.2 Materials (see 6.5).

3.2.1 Basic material. The basic material shall be cotton sateen cloth conforming to class 1 of MIL-C-10296, with the additional requirement that it show a minimum dynamic absorption of 60 percent when tested as specified in 4.3.1. The material shall be dyed Olive Green 107.

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3.2.2 Cuff tubing. The fabric for the cuff shall be a seamless, true one by one rib knit, dyed cotton tubing conforming to the following requirements when tested as specified in 4.3.1.

Yards per pound - minimum 9, maximum 12
Wales per inch, minimum - 16
Courses per inch, minimum - 20
Flat width, inches, minimum - 3

3.2.2.1 pH. The pH value of the water extracted from the finished cloth shall be not less than 5.0 or more than 7.0 when tested as specified in 4.3.1.

3.2.2.2 Color. The cloth shall be dyed Olive Green 109 and shall match the standard sample (see 6.3). The use of sulfur dyes and dyes containing elementary sulfur or components capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more labile sulfur than that shown by the standard sample when tested as specified in 4.3.1.

3.2.2.3 Matching. The color shall match the standard sample under artificial daylight having a color temperature of 7000 ± 500 K and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K.

3.2.2.4 Colorfastness. The dyed finished cloth shall show fastness to laundering, perspiration, and crocking equal to or better than the standard sample. When no standard sample is available, the dyed finished cloth shall show good fastness to laundering, perspiration, and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.3.1.

3.2.2.5 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material in the finished cloth shall not exceed 3.0 percent when tested as specified in 4.3.1.

3.2.3 Thread. The thread for seaming and stitching shall be cotton, type IA3, ticket No. 30 and 50, 3 ply, Olive Drab S-1, C.A. 66022, conforming to V-T-276.

3.2.3.1 Colorfastness. The dyed thread shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering (after 3 cycles).

3.2.4 Gimp. The reinforcement for buttonholes shall be cotton gimp thread, Olive Drab S-1, C.A. 66022, type I or II, size No. 8 conforming to V-T-280.

3.2.4.1 Colorfastness. The dyed gimp shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample. When no standard sample is available, the dyed gimp shall show "good" fastness to laundering (after 3 cycles).

3.2.5 Buttons. The buttons shall be type II, class D, style 26, 30-line conforming to V-B-871. The buttons shall have a dull finish and the color shall be Olive Green, BP, Cable No. 62016.

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3.2.6 Labels. Each liner shall have an identification label, a size label, and an instruction label, or a combination size, identification and instruction label, conforming to type VI of DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.2.6.1 Identification label. The identification label shall conform to class 1.

3.2.6.2 Size label. The size label shall conform to class 2, and shall contain the following information for each respective size. Descriptive information shall be printed in letters not less than a minimum 10 point.

X-small

Chest: Up to 33 inches
Stock No.:
Nato size: 7484

Small

Chest: From 33 to 37 inches
Stock No.:
Nato size: 8494

Medium

Chest: From 37 to 41 inches
Stock No.:
Nato size: 9404

Large

Chest: From 41 to 45 inches
Stock No.:
Nato size: 0414

X-large

Chest: From 45 inches and upward
Stock No.:
Nato size: 1424

3.2.6.3 Instruction label. The instruction label shall conform to class 3 and shall contain the following information:

LINER, SHIRT, CHEMICAL PROTECTIVE

1. Wear as a chemical protective layer when a chemical attack is imminent.

2. Wear over "T-shirt" or winter underwear.

3. Button flap on left side through right side buttonhole and then secure left side to the same button.

4. Tuck liner shirt into Liner, Trousers, Chemical Protective.

5. If necessary, use Buddy System when inserting hand into glove.

6. Be sure sleeve cuff of Chemical Protective Liner is held down at wrist.
Pull knitted cuff of glove over knitted cuff on liner and as high as possible up the sleeves.

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7. Launder by prescribed laundry methods. Retreat by Decontamination Kit M-13 Type or by Field Laundry Procedures.

8. Follow field laundry impregnation procedures strictly. Overheated clothing may ignite spontaneously.

DO NOT REMOVE THIS LABEL

3.2.6.4 Combination size, identification and instruction label. The combination label shall conform to class 14.

3.3 Design. The liner is a single breasted model having long sleeves and a stand-up collar. The liner has an attached protective flap with buttons to button along both front edges of liner.

3.4 Patterns. Standard patterns which provide a 3/8-inch seam allowance for all seams except double-lapped double-stitched seams for which a 1/2-inch seam allowance is provided, will be furnished by the Government. The standard patterns shall not be altered in any way, except as indicated by footnote to list of pattern parts, and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns.

3.4.1 List of pattern parts. The components of the liner shall be cut from the materials specified, according to the pattern parts indicated.

Material	Pattern nomenclature	Cut parts
Cloth, cotton, sateen	Front	2 1/
	Back	1
	Sleeve	2
	Collar	1
	Protective flap	1

1/ If not cut on selvage, provide sufficient seam allowance for turn under.

3.5 Construction.

3.5.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Where two or more methods for seam or stitch types are given for the same part of the operation any one of them may be used. When stitch type 401 is used, the looper (underthread) shall be on the inside of the liner. The gage of stitching for double-lapped, double-stitched seams shall be 1/4- to 5/16-inch gage, with the outer row of stitching 1/16 inch from turned edge on outside of liner. The gage of all overedge stitching, shall be 3/16-inch and the seaming row, when using safety stitching, shall be a 3/16-inch from back edge of overedge stitch. Seam allowance shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats or puckers shall result.

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* 3.5.1.1 Type 301 stitching. Ends of all stitchings shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

* 3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401, 503, 504, 515, 516, and 519 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.5.1.1.1 a. and 3.5.1.1.1 b.

* 3.5.2 Buttonholes. Buttonholes shall be the eyelet end, taper bar type, worked over gimp with the ends of the buttonhole stitching and gimp tacked with not less than 4 tacking stitching. The buttonholes shall be clean cut with the stitching securely caught in the fabric.

* 3.5.3 Bartack. Bartacks shall be 1/2 $\pm$ 1/16 inch long, 1/8 $\pm$ 1/32 inch wide and shall contain 28 stitches. The bartack shall be free from thread breaks and loose or tight stitching.

3.6 Manufacturing operations requirements. The liner shall be manufactured in accordance with operations specified in table I. The contractor is not required to follow the exact sequence of operations.

3.6.1 Figure. Figure 1 is furnished for information only. To the extent of any inconsistencies between the figure and the written specification, the written specification shall control.

3.6.2 Marking. The parts of the liner shall be marked to insure a uniform size throughout the liner. Any method of marking my be used except:

a. Metal fastening devices.
b. Sew-on type tickets.
c. Adhesive type tickets which leave traces of adhesive on the material upon removal of the tickets.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
	a. Cut liner in strict accordance with patterns furnished, which show size, warp directional lines, and notches for proper assembly of parts.						
	b. The filling effect side of the material shall be the face.						
2.	Replacement of defective components.						
	Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects or damages, as classified in 4.4.2.1 are excluded and replaced with nondefective and properly matched material.						
3.	Size marking (see 3.6.2).						
	Mark, ticket or bundle all parts of the liner.						
4.	Labels.						
	a. Sew the size, identification and the instruction labels, or combination size, identification, instruction label to inside center of back (1/2 inch tolerance) in the following arrangement with labels sewn on all four sides:	301	LSBj-1	10-14	30-3	30-3	

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labels. (cont'd)						
	(1) Size label - top edge caught in collar seam.						
	(2) Instruction label - top edge abutted with bottom edge of size label. The top edge may be caught in the bottom row of size label stitching.						
	(3) Identification label - top edge abutted with bottom edge of instruction label. The top edge may be caught in the bottom row of instruction label.						
	(4) Combination size, identification and instruction label - When a combination label is used it shall be positioned as in (1) above.						
5.	Make protective flap.						
	a. Fold flap as indicated by marks on pattern and stitch top and side.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Turn flap, force out edges and stitch 1/16 to 1/8 inch from edge at top and side, and with bottom raw edges turned in, continue stitching across bottom.	301 301	SSe-2(b) and SSc-1	10-14 10-14	30-3 30-3	30-3 30-3	
	or						
	c. With flap folded face out, turn in raw edges on top, side and bottom and stitch 1/16 to 1/8 inch from edge (may be precreased).	301	SSc-1	10-14	30-3	30-3	
	or						

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					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make protective flap. (cont'd)						
	d. Fold flap as indicated by marks on pattern and stitch top and bottom	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	30-3 30-3	30-3 50-3	
	and						
	e. Turn flap, force out edges and stitch top and bottom 1/16 to 1/8 inch from edge (see operation 11.b.).	301	SSe-2(b)	10-14	30-3	30-3	
	or						
	f. With flap folded face out, turn in raw edges at top and bottom and stitch 1/16 to 1/8 inch from edge (see operation 11.b.).	301	SSc-1	10-14	30-3	30-3	
6.	Seam facings on right and left front.						
	Fold facings to outside, as indicated by marks on pattern and stitch through front 1/16 to 1/8 inch from edge. (If facings are not cut on the selvage, turn in raw edge and stitch 1/16 to 1/8 inch from edge.)	301 or 301	EFa-1 or EFb-1	10-14 10-14	30-3 30-3	30-3 30-3	
7.	Join liner parts.						
	a. Join shoulder, sleeves to liner, and side and underarm seams with double-lapped double-stitched seams positioned as follows:	301 or 401	LSc-2 LSc-2	10-14 10-14	30-3 50-3	30-3 30-3	

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TABLE I.

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Join liner parts. (cont'd)						
	(1) Shoulder seams with backs over fronts.						
	(2) Sleeves to body with front and back parts over sleeves on outside of liner.						
	(3) Side seams and underarm seams in one continuous operation with fronts over backs.						
8.	Make and attach cuffs.						
	a. Cut cuff material of sufficient length to finish 6 ± 1/4 inches long. Fold to double thickness.						
	b. (1) Position top raw edges of cuff even with bottom edge of sleeve (sleeve inserted in cuff) and overedge stitch sleeve to cuff.	503 or 504	SSa-1 SSa-1	6-10 6-10	50-3 50-3	50-3 50-3	
	(2) Stitch cuff to sleeve 3/8 inch from edge.	301 or 401	LSq-2(a) LSq-2(a)	10-14 10-14	30-3 30-3	50-3 50-3	
	or						
	(3) Seam and overedge stitch cuff to sleeve in one operation.	515 or 516 or 519	LSbm-3(a)	10-14	30-3 (chain stitching) 50-3 (overedge stitching)	50-3 50-3	
	c. Turn cuff to finished position and raise stitch sleeve through seam allowance 3/16 to 5/16 inch from joining seam.	301 301	LSq-2(b) or LSbm-3(b) LSbm-3(b)	10-14 10-14	30-3 30-3	30-3 30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Hem bottom.						
	a. Turn up bottom of liner to outside, with the raw edge turned in and stitch 1/16 to 1/8 inch from edge, with the stitching continued across ends of hem 1/16 to 1/8 inch from edge. Hem shall measure finished 5/8 ± 1/8 inch wide.	301	EFb-1	10-14	30-3	30-3	
10.	Make and attach collar.						
	a. Fold collar and seam curved ends. Turn and force out edges.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Attach under collar to neck edge of liner.	301 or 401	SSa-1 SSa-1	10-14 10-14	30-3 30-3	30-3 50-3	
	c. Turn in bottom edge of top collar and stitch to liner 1/16 to 1/8 inch from turned edge and continue the stitching along outside edges of collar 1/16 to 1/8 inch from edges. The top edge of size label shall be caught in the top collar stitching (see operation 4.a(1)).	301 301 301	LSd-1 and SSe-2(b) and OSf-1	10-14 10-14 10-14	30-3 30-3 30-3	30-3 30-3 30-3	
11.	Attach protective flap.						
	a. Position protective flap on inside of left front as indicated by marks on pattern and stitch back (seamed) edge of flap through front 1/8 to 1/4 inch from edge. or	301	LSa-1	10-14	30-3	30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Attach protective flap. (cont'd)						
	b. If option in 5(d) and (e) or (f) is exercised, position raw side edge of flap on inside of left front, as indicated by marks on pattern, with fold toward side seam and seam. Turn flap toward front and raise stitch 3/16 to 1/4 inch from edge.	301	LSb1-2	10-14	30-3	30-3	
	c. Bartack through flap and liner, diagonally across stitching at top of flap.	Bartack		28 per bartack	30-3	30-3	
12.	Mark and make buttonhole.						
	Finished appearance. Buttonholes shall be horizontal with the eyelet end toward the front edge of liner with the inside edge 3/4 + 1/8 inch from front edges. The purling of the buttonholes shall be on the outside (facing side) of the liner.						
	Make eight 3/4 to 7/8-inch cut length buttonholes, four on each front, positioned as indicated by marks on pattern.	Button-hole		52-56 per button-hole including tack	30-3	50-3	
13.	Attach buttons.						
	Attach four buttons on protective flap, positioned as indicated by marks on pattern	101 or		14-16 per button	30-3		
		301		14-16 per button	30-3	30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Clean liner. a. Thread ends shall be trimmed to 1/4 inch maximum length. b. Remove loose thread ends. c. Remove spots and stains.						

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3.7 Finished measurements. The finished measurements shall conform to table II.

TABLE II. Finished measurements (in inches)

Sizes	1/2 Breast 1/	Back length 2/	Sleeve length 3/
X-small	20-3/4	31-5/8	21-3/4
Small	22-3/4	32-1/8	21-3/4
Medium	24-3/4	32-5/8	21-3/4
Large	26-3/4	33-1/8	21-3/4
X-large	28-3/4	33-5/8	21-3/4
Tolerances	± 3/4	± 1	± 1/2

1/ 1/2 breast - taken with liner buttoned, from folded edge to folded edge at base of armhole.

2/ Back length - taken at center back from bottom edge of collar to bottom edge of liner.

3/ Sleeve length - taken from base of armhole to bottom edge of cuff.

3.8 Workmanship. The finished liner shall conform to the quality of product established by this specification. The occurance of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirments as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition to testing provisions contained in component specifications, testing shall be performed on components listed in table III for characteristics noted. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. All requirements are applicable to the sample unit. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The unit for expressing lot size, and the sample unit for testing each component shall be in accordance with applicable subsidiary specifications and as follows:

Component	Lot size expressed as	Sample unit for testing
Cotton knit tubing	Yards	1-1/2 yards full width
Cotton sateen cloth	Yards	1/2 yard full width

TABLE III. Test methods

Component	Characteristic	Requirement paragraph	Test method
Cotton sateen cloth	Dynamic absorption	3.2.1	5500
Cotton knit tubing	Material identification	3.2.2	1200
	Type of knit	3.2.2	Visual 1/
	Yards per pound	3.2.2	5041
	Wales per inch	3.2.2	5070

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TABLE III. Test methods (cont'd)

Component	Characteristic	Requirement paragraph	Test method
Cotton knit tubing (cont'd)	Courses per inch	3.2.2	5070
	Width (flat)	3.2.2	5020
	pH	3.2.2.1	2811
	Color	3.2.2.2	Visual 1/
	Presence of labile sulfur	3.2.2.2	2020
	Colorfastness to:	3.2.2.4	
	- laundering		5610
	- perspiration		5680
	- crocking		5651
	Nonfibrous material Starch and protein content (including chloroform-soluble and water-soluble material)	3.2.2.5	2611

1/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".

4.3.2 End item examination. The end item shall be examined in accordance with 4.3.2.1 and 4.3.2.2. The lot size shall be expressed in units of one liner. The sample unit shall be one liner.

4.3.2.1 Visual examination. The end item shall be examined for the defects listed below. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 major, 15.0 major and minor A combined, and 40.0 total (major, minor A, and minor B combined) defects per hundred units.

Defect	Classification Major	Minor A	Minor B
Material defects and damages			
a. Applicable to cotton sateen:			
1. Any hole, needle chew, cut, tear, mend, or burn	X		
2. Any defect such as a smash, multiple float, or loose slub	X		

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Defect	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)			
b. Applicable to knitted cuffs:			
1. Hole, run, or dropped stitches	X		
2. Elasticity lacking, i.e., tight or stretched		X	
3. Slubby yarn			X
4. Mend			X
NOTE: Material defects and damages are to be classified as indicated above only when the condition is one which weakens the cloth or liner or when it is so conspicuously located as to be clearly noticeable. Non-weakening conditions which are inconspicuous shall not be classified as defects.			
Cleanness			
a. Any spot or stain on outside			X
b. Thread ends not trimmed to a maximum length of 1/4 inch or loose threads not removed from major portion of liner			X
c. One or more size markings not removed, when applicable			X
Components and assembly			
a. Any component part or required operation omitted (unless otherwise classified herein)	X		
b. Any operation not performed as specified (unless otherwise classified herein)		X	
Cutting			
Any component part not cut in conformance with directional lines on patterns or not cut in accordance with specification requirements	X		
Seams or stitching			
a. General:			
1. Any part of liner caught in an unrelated operation or stitching	X		
2. Seam twisted, puckered or pleated		X	
3. End of seam or stitching when not caught in other seam or stitching backtacked less than 1/4 inch			X
4. Thread break (all stitch types) stitched less than 1/2 inch beyond each end of break			X
5. Ends of continuous line of stitching not overlapped a minimum of 1/2 inch or as specified			X

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Defect	Classification Major	Minor A	Minor B
Seams or stitching (cont'd)			
b. Gage of stitching, edge or raised stitching:			
1. Unevenly gaged or various stitching not uniform			X
2. Beyond width specified or varies more than 1/16 inch when no range is specified			X
c. Open seams:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch to 1 inch inclusive		X	
- more than 1 inch	X		
NOTE: One or more broken or two or more continuous skipped or run-off stitches constitute an open seam. A double-stitched seam is considered open when either side of the seam is open.			
d. Raw edges:			
1. On outside:			
- up to 3/16 inch inclusive			X
- more than 3/16 inch	X		
2. On inside:			
- more than 1/4 inch		X	
NOTE: Raw edge not caught in stitching shall be classified as an open seam			
e. Run-offs:			
1. On joining seams, when resulting in an open seam, score as "open seam"			
2. On raised or top stitching when not resulting in an open seam (outside only):			
- 1/4 to 1/2 inch inclusive			X
- more than 1/2 inch		X	
3. On overedge stitching, more than 1/2 inch			X

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Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
f. Seam and stitch type:			
1. Not specified seam or stitch type		X	
2. Looper thread exposed on outside of liner when 401 stitch type is used	X		
g. Stitch tension:			
1. Loose tension, resulting in a loose seam	X		
2. Loose tension on edge or raised stitching resulting in a loosely exposed lower or top thread			X
3. Tight tension (stitches break when strain is applied in direction of seam)	X		
h. Stitches per inch (to be scored only when the condition exists on major portion of seam):			
1. Less than minimum specified:			
- one to three stitches			X
- four stitches		X	
- more than four stitches	X		
2. More than maximum specified:			X
i. Stitches skipped or broken on edge or raised stitching, when seam is seamed, turned and stitched:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch		X	
Buttonholes (construction)			
a. One or more omitted or added	X		
b. Stitches not caught in cloth, causing stitching to pull away from cloth	X		
c. Not specified type, or uncut buttonhole		X	
d. Ragged edges - one or more broken or skipped stitches in one or more buttonholes		X	
e. Gimp omitted:			
- in one buttonhole			X
- In two or more buttonholes		X	

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Defect	Classification Major	Minor A	Minor B
Buttonholes (construction) (cont'd)			
f. Purling on the inside of liner			X
g. Ends not tacked in:			
- one buttonhole		X	
- two or more buttonholes	X		
h. Cut length less than 3/4 or more than 7/8 inch		X	
Sleeves			
a. Cuff finished less than 5-3/4 or more than 6-1/4 inches long		X	
b. Cuff badly twisted		X	
c. Cuff spliced - not tubular construction		X	
d. Cuff not constructed of double thickness	X		
Buttons			
Missing, loose or defective:			
- one button		X	
- two or more buttons	X		
Bottom hem			
a. Formed on inside of liner			X
b. Stitching across end of hem omitted or insecure			X
c. Width less than 1/2 or more than 3/4 inch wide			X
Shoulder, armhole, underarm and side and sleeve seam			
a. Front lapped on back at shoulder seam			X
b. Back lapped on front at side seam and underarm			X
c. Sleeve lapped on front or back at armhole seam		X	
Front facings			
a. Stitched on inside of liner	X		
b. Back edge not turned under in stitching, when not cut on selvage		X	
c. Tight, short or twisted, causing excessive fullness or twist on front			X

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Defect	Classification Major	Minor A	Minor B
Front buttonholes (position)			
a. Unequally spaced by more than 1/2 inch	X		
b. Less than 5/8 or more than 7/8 inch from front edge		X	
c. Constructed in vertical position		X	
d. Eyelet end toward side seam			X
Protective flap			
a. One or more buttons out of alignment with front buttonholes by more than 1/4 inch		X	
b. Bartack at top edge missing or insecure	X		
c. Bartack misplaced			X
d. Bartack less than 3/8 or more than 1/2 inch long			X
Labels			
a. Size label, instruction label, or combination label missing, incorrect, or illegible	X		
b. Identification label missing, incorrect, or illegible			X
c. Not positioned as specified or not stitched on all four sides			X
d. Positioned off center of back by more than 1/2 inch			X

4.3.2.2 Finished measurements (see 3.7). Any measurement deviating from nominal dimension and tolerance in table of measurement or sleeves uneven in length by 3/4 inch or more shall be scored. The inspection level shall be S-3 and the AQL shall be 4.0 defects per hundred units.

4.3.2.3 Acceptable quality levels and inspection labels. The acceptable quality levels (AQL's) expressed in defects per hundred units (DHU) and inspection levels shall be as follows:

	AQL's	Inspection level
For 4.3.2.1:		II
Major	2.5	
Major and Minor A	15.0	
Two Major and Minor A and Minor B (combined)	40.0	
For 4.3.2.2:		S-3
One class	4.0	

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4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling Bulged or distorted container Improperly folded liners, inadequately tied bundles, contents improperly covered with paper
Content	Number of bundles is more or less than required Number of liners per bundle is more or less than required 1/ Size shown on one or more liners not as specified on container 1/

1/ For this defect, one bundle from each container in the sample shall be examined.

5. PACKAGING

* 5.1 Preservation. Preservation-packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each unbuttoned liner shall be laid out flat with the front up. The front width shall be adjusted to approximately 23 inches by over-lapping the left side on the right side. The sleeves shall be folded one over the other across the chest. The bottom edge shall be brought up toward the shoulder line (for the larger sizes it may be necessary to prefold the bottom edge), so that the completely folded liner shall measure approximately 23 by 14-1/2 inches. Ten folded liners of one size only, each alternately reversed end for end, shall be neatly stacked in a bundle. Each bundle shall be securely tied with cotton tape or twine approximately four inches in from each end.

5.1.2 Commercial. The liners shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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* 5.2.1 Level A packing. Forty liners of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container, conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bundles shall be packed flat four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

* 5.2.2 Level B packing. Forty liners of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A bundles shall be packed flat four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II, as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

* 5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.

* 5.2.3 Commercial packing. Liners, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

* 5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side, directly under the printing or stencilling, a white paper label 5 by 4 inches, with the words "MIXED SIZES" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity and size contained therein.

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6. NOTES

6.1 Intended use. The liner is intended for wear by personnel of the Department of Defense as an inner garment to provide protection against chemical warfare agents.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to sample of the end item, address the procuring activity issuing the invitation for bids.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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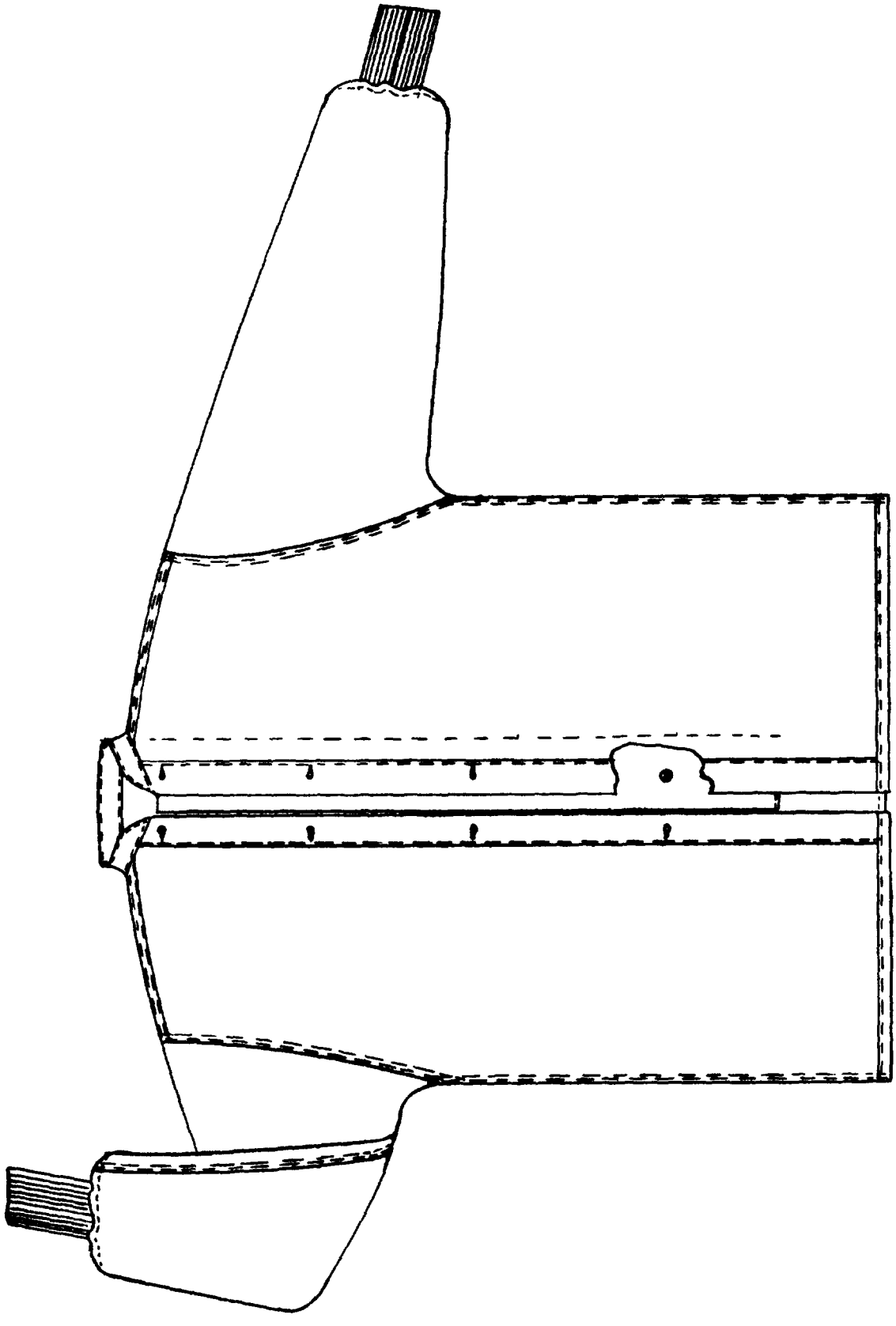


FIG. 1 LINER, SHIRT, CHEMICAL PROTECTIVE 2-1-8129

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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